

Project 2: Boolean Arithmetic

The centerpiece of any computer's architecture is its CPU, or Central Processing Unit, and the computational centerpiece of the CPU is the ALU, or Arithmetic-Logic Unit. In this project you will gradually build a set of chips that carry out arithmetic addition, culminating in the construction of the ALU chip of the Hack computer. In the process, you will witness, hands-on, how arithmetic operations on signed numbers can be realized using Boolean logic.

Objective

Build the following chips:

HalfAdder

FullAdder

Add16

Inc16

ALU

We note in passing that the chips listed above (as well as their names) are standard, except for the ALU, which varies from one computer architecture to another.

Implementation

As in project 1, building the chips requires tools, guidelines, and tips. All these resources are described in the [implementation notes](#).

Project-specific Tips

The Hack ALU computes a 16-bit value named out. In addition, it computes two 1-bit outputs named zr and ng. We recommend building the ALU in two stages. First, implement a basic ALU that computes out, ignoring the zr and ng outputs. Then implement a final version that computes out as well as zr and ng. We provide two sets of files for testing each stage: ALU-basic.tst and ALU-basic.cmp for testing the basic version; ALU.tst and ALU.cmp for testing the final version.

If you are using the online simulator, which is the preferred option, you can select the ALU test script from the Test drop-down menu; If you are using the legacy desktop simulator, you can load the ALU test script from the projects/2 folder.